

FIFA World Cup 2026 as a Contamination-Free Benchmark for LLM Forecasting Agents: Four Models, a Bookmaker, and 104 Matches

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Abstract

We introduce **WC2026-Agents**, a benchmark and dataset for evaluating large language models (LLMs) as *autonomous forecasting agents* on real, future events. For every one of the 104 matches of the 2026 FIFA World Cup, four frontier models, namely Claude Opus 4.8, ChatGPT (GPT-5.5, high reasoning), Gemini 3.1 Pro, and Grok (Expert Mode), ran an identical search-act-reflect loop: gather evidence with a web tool, commit to a 1X2 (team-A win / draw / team-B win) probability distribution and a virtual \$100 bet, and, after the match, reflect given only the final score. Because every match kicked off after the models' training cutoffs, the benchmark is *contamination-free* by construction. Crucially, we pair the four agents with a fifth competitor drawn from the same information environment, the pre-match betting market, collected as per-match 1X2 odds, giving an economically grounded human baseline and letting us score not just what an agent predicts but what it *does with money*. The release contains 416 forecasts and 414 reflections with verbatim reasoning, ground truth (including penalty shootouts), odds, and a reproducible evaluation suite. A reference evaluation surfaces findings that raw accuracy hides: the four agents issue an identical top pick in 92% of matches and none beats the market's Brier score, yet they diverge sharply as decision-makers: betting return-on-investment ranges from -18% to $+10\%$, fading the market lowers every agent's hit rate, the share of forecasts that cite the market ranges from 12% to 100%, and self-reported error rates on wrong picks range from 36% to 86%. The benchmark thus measures calibration, decision quality, and self-knowledge, the axes on which frontier models differ even when their predictions do not. Data and code: <https://github.com/graphuofm/FIFA2026LLM>.

CCS Concepts

• Computing methodologies → Artificial intelligence; • Information systems → Data mining.

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Keywords

large language models, LLM agents, forecasting, calibration, prediction markets, sports analytics, benchmark dataset, data contamination

1 Introduction

Large language models are moving from answering questions to *acting*: retrieving evidence, reasoning, committing to decisions, and reflecting on outcomes [22, 34, 36]. Evaluating this agentic behaviour is hard for three reasons. First, most benchmarks are static and increasingly *contaminated*: test items leak into training data [8, 17]. Second, they score a single answer and ignore what an agent would *do* with a belief. Third, they lack a strong, economically meaningful human baseline.

Real-world forecasting of *future* events dissolves all three problems at once. A scheduled sporting tournament provides a stream of questions whose answers (i) do not exist at query time, so contamination is impossible; (ii) arrive on a known date with unambiguous ground truth; (iii) are continuously priced by a liquid betting market that aggregates human and algorithmic beliefs into calibrated probabilities [28, 31]. The 2026 FIFA World Cup, with 104 matches over five weeks, every one after current models' training cutoffs, is close to an ideal such stream.

We use it to build **WC2026-Agents**, a benchmark that treats four frontier LLMs as forecasting agents and the betting market as a fifth competitor. For each match, each agent runs a fixed *search-act-reflect* loop: it searches the web for team news and prices, acts by returning calibrated 1X2 (team-A win / draw / team-B win) probabilities together with a virtual \$100 staking decision, and, once the score is known, reflects on its own forecast. We settle every bet at real 1X2 odds, so the benchmark scores three distinct capabilities that usually travel together but here come apart: *predictive calibration* (are the probabilities right?), *decision quality* (does the agent profit against a real price?), and *self-knowledge* (does it know when it was wrong?).

A reference evaluation of the four 2026 models illustrates why these axes matter. On *what will happen*, the agents are nearly interchangeable: they choose the same most-likely outcome in 92% of matches, their accuracies fall in a three-point band, and none beats the market's Brier score, an efficient-market result that a single-answer benchmark would report as a null. On *what to do about it*, they diverge sharply: betting ROI spans -18% to $+10\%$; fading the market lowers every agent's hit rate and loses money for three of the four; the fraction of forecasts that even mention the market spans 12% to 100%; and, on their own wrong picks, self-reported

error rates span 36% to 86%. These behavioural axes, not accuracy, are where frontier models differ.

Contributions.

- **A contamination-free agentic-forecasting benchmark** over all 104 matches of a future World Cup: 416 forecasts and 414 reflections from four frontier LLMs, with verbatim reasoning, paired pre-/post-match structure, and a reproducible evaluation suite (§4, §5).
- **A market baseline as a fifth competitor.** We release per-match 1X2 odds and the settlement machinery so any forecaster can be scored not only for calibration but for realized betting profit against a real price (§4.5).
- **A curated, documented dataset.** Fixtures, team orientation, scores (including penalty shootouts and who advanced) were hand-resolved from four heterogeneous transcripts and cross-checked four ways (§4.4).
- **A reference evaluation** spanning calibration, betting/decision quality, reasoning content, and reflection honesty, with the findings above and a distillation of the agents' recurring blind spots into candidate features (§6).

2 Related Work

LLMs as forecasters. Retrieval-augmented LLMs can approach human crowd accuracy on judgemental questions [11], and LLM ensembles rival human crowds [23], echoing the human superforecaster literature [26]. Dynamic forecasting benchmarks such as FORECASTBENCH [15] and early neural event forecasting [37] score probabilistic accuracy. We add three elements those lack: a fixed agentic loop, a betting-based *decision* score against a real market, and paired self-reflection.

LLM agents, reasoning, and reflection. Agentic scaffolds interleave reasoning and tool use [30, 36] and let models critique and revise their own outputs [18, 24], though unaided self-correction is fragile [13]. Surveys chart the space [29, 34]. Our reflection prompt is a minimal, standardized probe of self-knowledge at tournament scale.

Calibration and uncertainty. Modern networks are often miscalibrated [10]; LLM confidence can be elicited but is imperfectly calibrated [14, 27, 35]. We measure calibration with reliability curves and expected calibration error [20], and full-distribution quality with the Brier score [4] and log-loss.

Betting and prediction markets. Bookmaker odds are strong, hard-to-beat probability estimates [1, 28, 32], consistent with market efficiency [7] and exhibiting known regularities such as the favorite-longshot bias [16, 25]. Machine-learning football models are a long line of work [3, 5]. We use the market both as a baseline and as the price an agent may follow or fade.

Sports as a forecasting domain. Football pairs a strong, liquid pricing signal with irreducible low-scoring variance, so a favourite's edge is genuine yet rarely decisive; a well-judged 60/25/15 is a better answer than a confident single label, which makes the sport a probabilistic rather than a deterministic prediction task. Its three-way structure also forces a forecaster to place mass on the draw, an outcome that books price with care and that naive predictors systematically underweight [5]. We exploit both properties: the

draw makes calibration non-trivial, and the liquid market gives every match a reference price against which an agent can be scored and can trade. Unlike closed-form statistical models fit to historical results, our agents forecast strictly future fixtures from live web evidence, so the domain's variance is faced online rather than in cross-validation.

Benchmarks and contamination. Static benchmarks [12, 17] face contamination as models train on ever more of the web [8]. Forecasting strictly future events is contamination-free by construction, a property we exploit. Finally, algorithm aversion [6] and overconfidence [19] frame our behavioural analysis.

3 Task Formulation

A match x has an outcome $o(x) \in \mathcal{Y} = \{\text{team_a_win}, \text{draw}, \text{team_b_win}\}$, where team A is the first-listed side. A forecasting agent g observes x and its public information environment and returns a distribution $p_g(x) \in \Delta(\mathcal{Y})$ and a decision $d_g(x) = (\text{pick}, \text{stake})$ with stake $\in [0, 100]$. After the match it returns a reflection $r_g(x)$. The market m supplies an implied distribution $p_m(x)$ obtained by removing the vigorish from published 1X2 odds; it makes no bets and files no reflection, serving as a fifth forecaster and as the price at which agent bets settle. Knockout ties that finish level go to a shootout; because a 1X2 bet settles on the 90-minute result, we take $o(x)$ to be that result (a draw for such ties) and separately record the shootout winner (§4.4).

4 The WC2026-Agents Benchmark

4.1 Agent protocol

Each match generates two turns in a single per-model conversation. The *pre-match* turn ($T-24\text{h}$) instructs the agent that it is in a research study (virtual money; do not refuse), names the fixture, tells it to use web search, and requires a strict JSON reply:

```
{"probabilities": {"team_a_win", "draw",
"team_b_win"},
"bet": {"pick", "stake_usd"},
"reasoning": ..., "key_sources": [urls]}
```

The *post-match* turn provides only the final score and asks, without web access, for a reflection: an outcome_vs_prediction label (correct/partially_correct/incorrect), a calibration self-judgement, a luck attribution, one key_factor_i_missed, one key_factor_i_overweighted, a counterfactual, a would-bet-differently verdict, and a confidence. The prompt text is byte-identical across agents; only the model changes.

4.2 Models

We evaluate four frontier assistants queried through their consumer interfaces with web tools enabled and held fixed across the tournament: Claude Opus 4.8 [2], ChatGPT (GPT-5.5 "Thinking", high reasoning effort) [21], Gemini 3.1 Pro [9], and Grok in Expert Mode [33]. Using production assistants (rather than raw API base models) is deliberate: it captures the retrieval-and-reason *agent* a user actually deploys, which is the object of study.

4.3 Data collection

Between 11 June and 19 July 2026, each agent was run on all 104 fixtures (*m01–m72* group, *m73–m104* knockout), producing a pre-match forecast ~24h before kickoff and a post-match reflection after the result. Raw transcripts are archived read-only. The pre-match window strictly precedes each outcome, so no target could appear in any model’s training data.

4.4 Curation and quality control

Group-stage transcripts follow a regular structure and were parsed with a string-aware JSON extractor. The knockout transcripts are not regex-parseable: they use four different header renderings, some Round-of-16 fixtures were pasted with bracket placeholders (“W74 vs W77”) whose numbering does not match reality, and final scores appear in a dozen phrasings (penalty shootouts, lower-cased names, occasionally reversed team order). We therefore *hand-resolved* fixtures, team orientation, and ground truth for all 32 knockout matches from the transcripts, encoding them in a verified table. As an automatic guard, the human-pasted score line in each of the four agents’ transcripts is compared to that table; the four-way check flagged zero disagreements across all 128 knockout cells. Ground truth records the 90-minute result, the shootout result where applicable, and which side advanced. Every deviation from the JSON schema (recovered braces, duplicated pastes, two missing Gemini reflections) is logged rather than dropped.

4.5 Market odds

We release pre-match 1X2 odds for all 104 matches with a source URL each, oriented to team A and vig-removed to implied probabilities (mean overround 1.05). Knockout odds are opening lines from public sportsbook previews; we validated them against the market probabilities the agents themselves cited in their reasoning (e.g. our de-vigged Brazil–Japan line, 55/25/20, matches a cited “56/25/19”), and corrected one line whose implied hold was economically impossible. The market is thus both an independent baseline and the settlement price for agent bets.

4.6 Schema, statistics, accessibility

The release is one row per (agent, match): fixture and stage; ground truth with penalty and advancement fields; the pre-match probabilities, pick, stake, verbatim reasoning and cited URLs; the eight reflection fields; and parse-issue flags. A tidy 104-match table joins all agents and the market for analysis. Table 1 summarizes coverage. All data (CC BY 4.0) and code (MIT), including the one-command pipeline that regenerates every table and figure, are public; the dataset requires no access request.

An illustrative record. For match *m92* (Brazil vs. Norway, Round of 16), one agent returned the pre-match forecast {team_a_win: 0.55, draw: 0.25, team_b_win: 0.20}, a \$40 stake on Brazil, and the reasoning “Brazil enters as heavy favorites with superior squad depth . . . Norway’s Haaland–Ødegaard threat and historical edge make an upset possible.” Norway won 2–1. The paired reflection then labelled the outcome incorrect, named the missed factor as “Norway’s historical dominance and counter-attacking effectiveness . . . amplified by Haaland,” and the over-weighted factor as “Brazil’s

Table 1: Dataset coverage. Two Gemini group reflections are absent in the source and flagged, not imputed.

Stage	Matches	Agents	Forecasts	Reflections	Bets placed
Group	72	4	288	286	232
Knockout	32	4	128	128	103
Total	104	4	416	414	335

Table 2: Task T1: calibrated prediction over all 104 matches. Best in bold; $\uparrow/\downarrow$ = higher/lower better. The market attains the best Brier score.

	Acc. $\uparrow$	Brier $\downarrow$	LogLoss $\downarrow$	ECE $\downarrow$	Draw mass
Claude	.664	.4705	.806	.116	.234
ChatGPT	.683	.4729	.814	.111	.235
Gemini	.654	.4828	.820	.068	.238
Grok	.683	.4706	.803	.097	.227
Market	.683	.4688	.807	n/a	.229

overall squad depth and pedigree.” Every record has this pre/post structure, letting a user trace a belief, a decision, a result, and a self-assessment for the same event.

5 Benchmark Tasks and Metrics

The suite scores three capabilities.

(T1) Calibrated prediction. Accuracy ($\arg \max p_g = o$); multi-class Brier $\sum_i (p_{g,i} - \mathbb{1}[o=i])^2$; log-loss $-\log p_{g,o}$; and, for the top pick, a reliability curve and expected calibration error (ECE) [20] over ten confidence bins. *Convergence*, the fraction of matches on which all agents share an arg max, quantifies how much accuracy can separate them.

(T2) Decision quality. Each agent’s own pick/stake settles at the match’s decimal odds on the 90-minute result: profit = stake $\times$ (dec – 1) on a win, else –stake. We report bankroll, ROI, hit-rate, and a *contrarian* split: a bet is contrarian if its pick differs from the market’s most-likely outcome, isolating the value of fading the price. The market’s implied probabilities are scored under T1 as a fifth forecaster.

(T3) Behavioural transparency. Every reasoning and reflection field is coded into 13 factor categories with a published keyword lexicon (no model in the loop), giving each agent’s information diet and its *self-knowledge*: on the matches its arg max got wrong, how often does its reflection label itself incorrect?

6 Reference Evaluation

We evaluate the four 2026 agents and the market. Tables 2 and 3 give the headline numbers; figures expand each result.

6.1 The agents converge; accuracy cannot separate them

The four agents issue an identical top pick in **92%** of matches (91.7% group, 93.8% knockout; Fig. 1). Pairwise agreement reaches 100% (ChatGPT–Grok) and never falls below 92%. Accuracy therefore

Table 3: Task T2: betting/decision quality over all 104 matches. Each agent settles its own picks at real 1X2 odds. “Contr.” = contrarian (against-market) bets.

	#Bets	Staked	Net \$	ROI %	Hit	Contr. share/ROI
Claude	73	1519	−275	−18.1	.329	.58 / −24%
ChatGPT	55	1471	+118	+8.0	.509	.36 / −14%
Gemini	103	8660	+322	+3.7	.631	.14 / −12%
Grok	104	6305	+650	+10.3	.673	.05 / +33%

spans only three points (Table 2), i.e. three matches. Figure 2 shows the mechanism: each agent’s team-A probability tracks the market’s almost perfectly ($r = 0.97\text{--}0.99$), so the agents are, to first order, reading and lightly re-expressing the same price. On the eight matches where they are not unanimous, the split is always 3–1 (a single agent dissents), and the dissenter is usually Gemini (six of eight), consistent with it being the least market-anchored agent (Fig. 6). All eight are near-coin-flips where the market itself is close to even, so a shade more weight on one side tips the arg max. Disagreement thus appears exactly where the signal is weakest, not where an agent has found private information.

6.2 No agent beats the market

Scored as a forecaster, the market attains the best Brier score of all five competitors (0.469 vs. 0.471 for the best agent) and ties the best accuracy (Table 2; per-stage in Table 4). It also identifies the advancing side in 24 of 32 knockout ties (75%), equal to the agents. This is the expected efficient-market outcome [28]: web-enabled agents recover the market’s information but do not systematically exceed it. The tournament was mildly favorite-friendly, which the released odds let users control for: binning matches by the market’s implied probability for its favored side, that side won 62% of the time when priced near 48% and 78% when priced near 62%, while heavy favorites (~81%) were well-calibrated. In a cup with slightly fewer upsets than priced, every forecaster that respected the market, agents and market alike, looks under-confident in hindsight (§6.3), a difficulty profile the benchmark exposes rather than hides. Per stage (Table 4), knockout matches are markedly more predictable for every competitor (accuracy .719 vs. ~.65; lower Brier), as seeding concentrates mismatches; the agents stay indistinguishable within each phase and the market again leads on Brier.

6.3 Calibration does separate them

All four agents lie *above* the reliability diagonal: they are **under-confident** on their top pick, which wins more often than its stated probability (Fig. 3). Gemini is best calibrated (ECE 0.068), then Grok (0.097), ChatGPT (0.111), Claude (0.116). Calibration and full distribution can disagree: Gemini has the best top-pick ECE but the worst Brier, because it spreads probability mass more aggressively and stakes larger (Table 3).

The four agents agree on 92% of picks

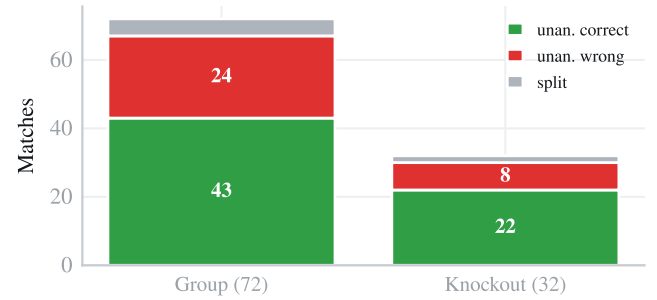
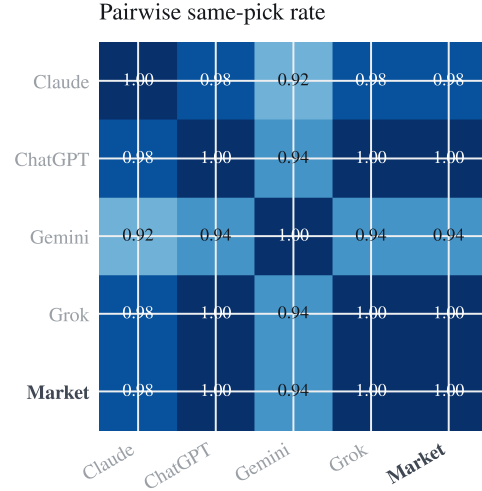


Figure 1: Convergence. Top: pairwise same-pick rate. Bottom: match counts by consensus; genuine disagreement is rare and unanimous picks are right about twice as often as wrong.

Table 4: Task T1 by phase (accuracy / Brier / log-loss). Group ($n=72$) and knockout ($n=32$).

	Group			Knockout		
	Acc	Brier	LogL	Acc	Brier	LogL
Claude	.639	.489	.828	.719	.428	.756
ChatGPT	.667	.493	.838	.719	.429	.759
Gemini	.625	.485	.820	.719	.477	.822
Grok	.667	.491	.826	.719	.425	.752
Market	.667	.486	.827	.719	.431	.761

6.4 The draw is where the three-way structure bites

Nearly a quarter of matches (24 of 104) ended level after 90 minutes, including the four knockout ties later settled on penalties. In aggregate the agents price this well: each assigns 23–24% mean mass to

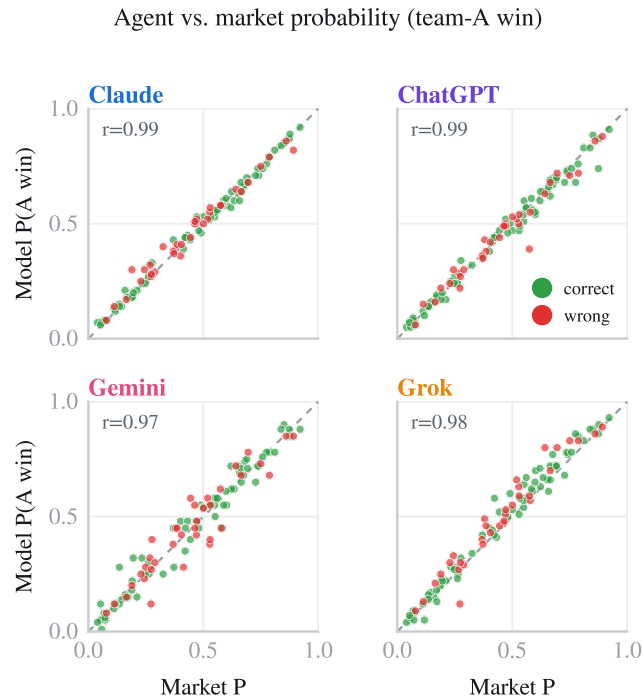


Figure 2: Each agent’s probability for a team-A win against the market-implied probability, all 104 matches. Points hug the diagonal (r shown); Gemini is the loosest, consistent with it rarely citing the market (Fig. 6).

the draw (Table 2, last column), almost exactly the empirical 23%. Yet they almost never *pick* the draw, because it is rarely any single match’s arg max; the mass is spread thinly across fixtures rather than concentrated on the ones that actually finish level. Draws are thus the largest single source of Brier loss: a level result is one on which every agent that committed to a side is wrong at once: of the 32 matches missed by all four agents, 22 were 90-minute draws, including four of the eight unanimous knockout misses (Fig. 8). The market carries marginally more draw mass (0.229) and places it more accurately, one concrete reason it edges the agents on Brier while only tying them on accuracy. For benchmark users this isolates a separable, measurable weakness, draw calibration as distinct from favourite and underdog calibration, that can be re-scored directly from the released distributions.

6.5 Decision quality diverges where accuracy does not

Settling each agent’s own picks at real odds, betting ROI spans -18.1% (Claude) to $+10.3\%$ (Grok), with net results of $-\$275$ to $+\$650$ (Table 3, Fig. 4). The spread is not driven by who they pick, since they pick alike, but by *how much they stake and whether they respect the price*. Gemini stakes most aggressively ($\$8,660$ across 103 bets); Claude stakes least and most often on draws and underdogs. As a reference, a naive baseline that stakes a flat $\$100$ on the market favourite every match returns $+\$1,041$ (grey line, Fig. 4), out-earning all four agents in absolute terms at comparable ROI,

Reliability of the top pick — colour = agent, grey = market (all under-confident)

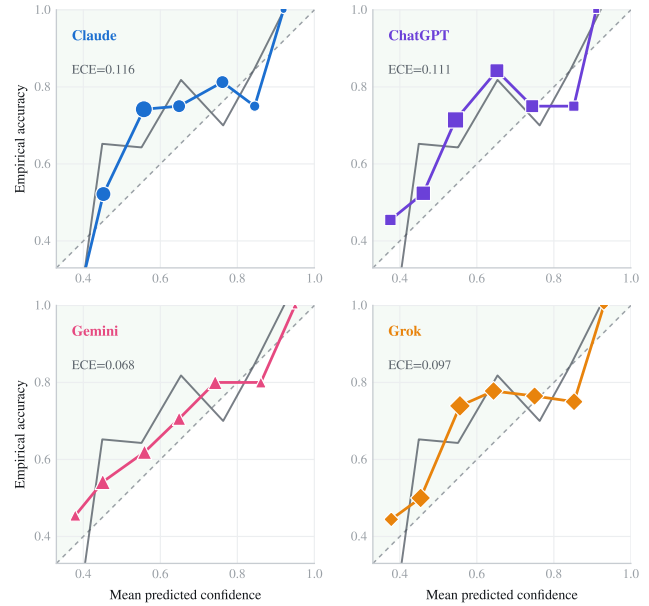


Figure 3: Reliability of each agent’s top pick, with the market as a grey reference curve. All lie in the shaded (under-confident) region; marker area $\propto$ bin count. ECE also appears in Table 2.

a betting counterpart of the result that no agent beats the market. (Favorites over-performed their price this cup, $\$6.3$; the released odds let users hold this difficulty fixed.)

6.6 Fading the market rarely pays

Decomposing bets by their relationship to the price explains the P&L (Fig. 5). Contrarian (against-market) bets hit 21–40% and return -24% to $+33\%$ (the lone positive being Grok’s five such bets), versus 48–69% for market-conforming bets, which supply nearly all of the profit (Claude’s conforming bets also lose, $-\$108$). The ordering is monotone: the more an agent fades the market, the lower its net profit. Claude cites the market in every forecast yet bets against it 58% of the time (it *sees* the price and overrides it) and is the only agent with a net loss.

6.7 Staking discipline is a distinct axis

Beyond *which* side to back, agents differ in how they *size* a bet. We correlate each agent’s stake with the edge it believed it held (its probability minus the market’s for the picked side) and with its confidence. Grok’s stake tracks its confidence most tightly (Pearson $r=0.85$) and its claimed edge positively ($r=0.13$), a Kelly-like behaviour that concentrates money on its strongest reads. Gemini and ChatGPT also stake up with confidence ($r=0.54$ and 0.53), but Gemini’s mean stake ($\$84$ of a $\$100$ budget) is a near-max-bet habit that inflates variance, so its bankroll in Fig. 4 is the most volatile of the four. Claude is the reverse: its stake is *negatively* related to

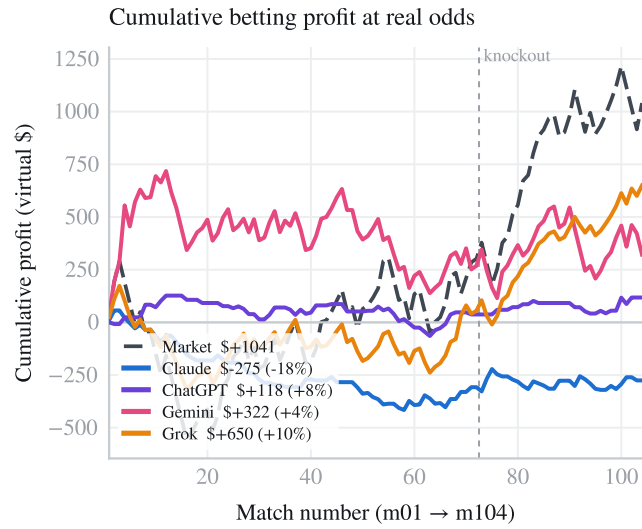


Figure 4: Cumulative virtual profit over the 104-match tournament, each agent settling its own stakes at real 1X2 odds. Trajectories diverge despite nearly identical point predictions.

confidence ($r = -0.20$), because it sizes up precisely on the low-confidence, against-market “value” plays that tend to lose. Staking discipline is therefore a separate, measurable axis, only loosely coupled to whether a pick is correct, and the released stakes let others re-score the agents under any fixed staking policy.

6.8 Information diets differ while picks converge

Coding the reasoning reveals sharply different information diets that nonetheless yield the same picks (Fig. 6). The share of forecasts that cite the betting market or odds ranges from **100% (Claude)** and **92% (ChatGPT)** to **12% (Gemini)**; Gemini reasons almost entirely from football (attacking threat, defensive organisation, form) while barely referencing the price. This is the behavioural counterpart of Fig. 2: Gemini is market-blind in its *stated reasoning* yet still lands near the market, and consequently fades it in only 14% of its *bets*. Grok most often invokes “paper strength” (squad depth 88%, form 76%, FIFA ranking 61%); Claude most often invokes knockout variance (53%).

6.9 Reflection is a behavioural fingerprint

Given only the final score, the agents reflect very differently on their own mistakes (Fig. 7). On the picks each got wrong, the rate of admitting “incorrect” ranges from **86% (Gemini)** to **36% (ChatGPT)**, which softens 55% of its errors to “partially correct”; Claude (49%) and Grok (61%) sit between. When they were right, the rate of claiming “correct” ranges from 68% (Claude) to 94% (Grok). Reflection style is thus a stable per-agent trait, orthogonal to accuracy, and relevant wherever a model’s self-report gates a downstream action [13, 24].

Contrarian bets win less often for every agent

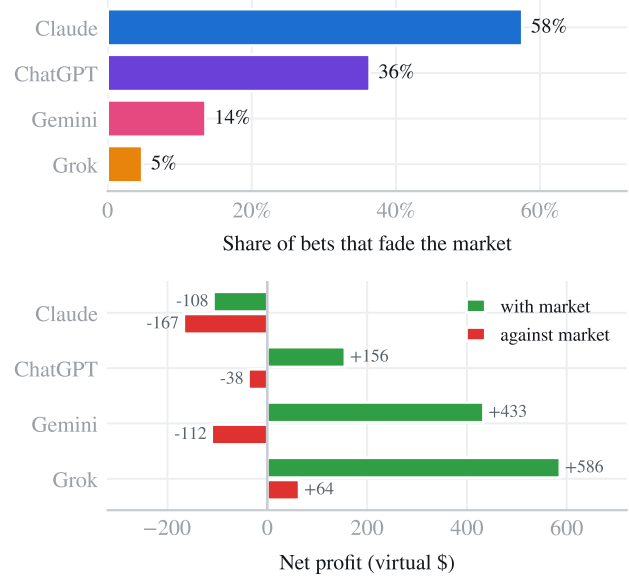


Figure 5: Against-the-market analysis. Top: share of each agent’s bets that faded the market favourite. Bottom: net profit from market-conforming vs. contrarian bets. Contrarian bets hit less often than conforming ones for every agent and lose money for all but Grok (five bets, +\$64).

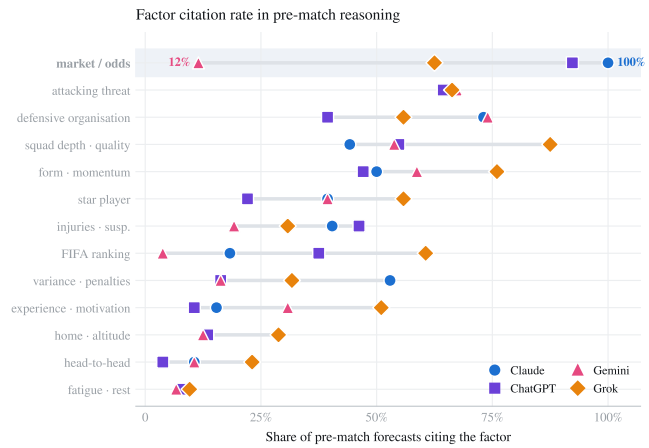


Figure 6: Share of pre-match forecasts citing each factor, per agent (dumbbell: one marker per agent). Market/odds citation ranges from 12% to 100%.

6.10 Shared blind spots and candidate features

All four agents were wrong on 32 of the 104 matches (24 group, 8 knockout), 22 of them 90-minute draws. The eight knockout matches mispredicted by *all four* agents (Fig. 8) are the sharpest cases: in every one they backed the favourite. Four were penalty

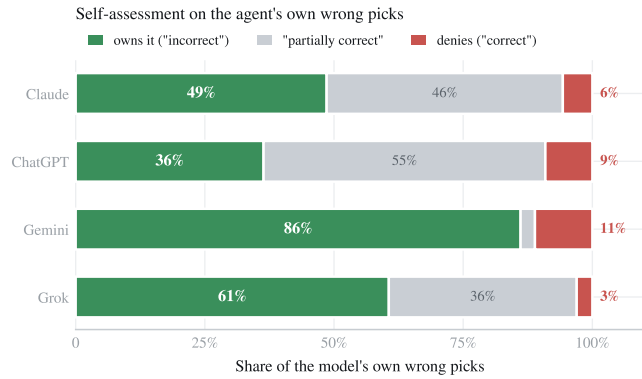


Figure 7: Self-assessment on each agent's own wrong picks. Owning error ("incorrect") ranges from 36% to 86% across agents.

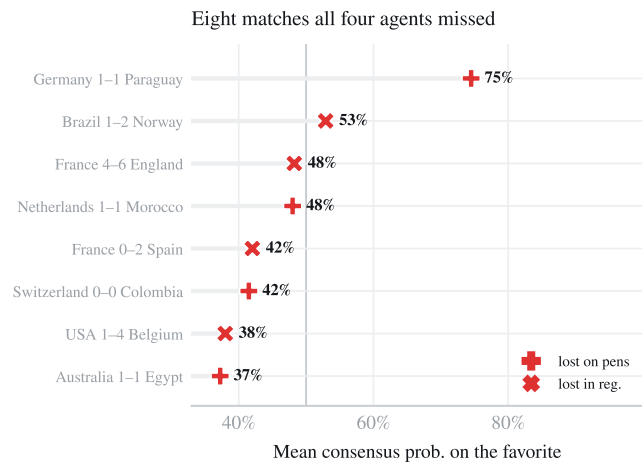


Figure 8: The eight knockout matches mispredicted by all four agents. Every agent backed the favourite; four ties were lost on penalties after a 90-minute draw.

ties in which a heavy favourite (Germany, 75% consensus) failed to win in 90 minutes; the rest were regulation upsets (e.g. Norway over Brazil, Belgium 4-1 over the USA, Spain over France). The agents' own post-mortems are strikingly consistent: across all wrong picks, the factor most often named as *missed* is underdog defensive organisation (62%), then knockout/penalty variance (32%); the factor most often named as *over-weighted* is the favourite's squad quality (40%) and attacking firepower (38%). These yield concrete, minable priors for future systems: down-weight a favourite's paper talent against a disciplined low block, and explicitly price the draw→shootout path in knockout ties.

Three of these priors are directly minable from the release and target the recurring errors above. (1) A *low-block indicator*: how often the underdog's recent matches were low-scoring and defensively compact, the factor the agents' own post-mortems name most often as missed (62%). (2) A *shootout-risk prior* for knockout ties that lifts draw and upset mass when a heavy favourite meets

a compact opponent, since four of the eight unanimous knockout misses were penalty ties. (3) A *market-deviation gate* that flags any forecast departing from the price without a stated non-price rationale, the exact configuration under which contrarian bets hit less often for every agent and lost money for three of the four (Fig. 5). Each is a single transformation on the released reasoning, odds, and outcome fields and is scorable against the same 104-match ground truth, so the benchmark supplies not only a diagnosis but a concrete, testable next step.

7 Discussion and Impact

The benchmark separates three capabilities that a single-answer score conflates. On prediction the 2026 agents are interchangeable and no better than the market; on *decision* and *self-knowledge* they differ by large, stable margins. For practitioners the lesson is that selecting an LLM for a consequential forecasting-and-acting pipeline should weigh calibration, staking discipline, and honest self-assessment, not headline accuracy. Because the protocol is a thin wrapper around a consumer assistant, WC2026-Agents is a template: the same search-act-reflect loop and market settlement apply to any scheduled, market-priced event stream (other tournaments, elections, product launches), enabling live, rolling, contamination-free evaluation and longitudinal tracking of model releases. We release the loop, lexicon, and settlement code to that end.

Two results deserve emphasis for benchmark design. First, the convergence of four independently built models cautions against reading a single leaderboard number as a capability signal: when models share a retrieval surface they also share errors, so an ensemble inherits rather than cancels the common bias, a limit on the "wisdom of the silicon crowd" [23] when the crowd reads the same web. Second, that no agent beats a vig-bearing market is the sharpest available null for an over-eager "LLMs can predict the future" narrative: the market is not a weak baseline but a near-efficient [7, 28] aggregator that the agents match but do not exceed, so a benchmark that omits it would mistake market-level performance for super-human forecasting. Surfacing such failure modes is precisely the role of a datasets-and-benchmarks contribution.

A third lesson concerns *why* decision quality diverges when predictions do not. In our data, betting outcome is explained far less by which side an agent backs, since all four back the same side, than by two behavioural traits the benchmark isolates: how tightly an agent sizes its stake to its own confidence, and how often it overrides the market price. Disciplined, confidence-proportional staking paired with rare, selective contrarian bets ends the tournament ahead; staking up on low-confidence, against-market "value" plays ends it behind, despite near-identical point predictions. The operative capability for a deployed forecasting-and-acting system is therefore not a better belief but a better *policy over a shared belief*, a distinction a prediction-only benchmark cannot observe and one that grows more important as agents are handed real budgets and irreversible actions.

These distinctions translate into concrete guidance. A team that needs only a *probability*, for example to populate a dashboard or seed a simulation, can treat the 2026 agents as interchangeable and should simply defer to, or lightly ensemble with, the market. A team that will *act* on the forecast, by allocating budget, triggering

a workflow, or gating a decision, should instead evaluate on T2 and T3: prefer the agent whose staking is confidence-proportional and whose self-reports track its actual error rate, and read frequent market-fading as a warning sign rather than a source of edge. Because all three tasks are scored from the same released records, a practitioner can reproduce this selection under their own risk profile rather than trusting a single leaderboard number.

8 Using WC2026-Agents

The release supports several lines of work out of the box, each a single join on the tidy 104-match table. **(i) Calibration and uncertainty.** 416 forecasts with confidences and outcomes, plus a market reference, form a contamination-free testbed for LLM (mis)calibration and confidence elicitation [27, 35]. **(ii) Agentic decision-making.** Paired probabilities, stakes, and real odds let researchers score arbitrary staking policies (flat, Kelly, learned) and quantify the value of following versus fading a price, a decision layer absent from prediction-only benchmarks [15]. **(iii) Self-knowledge.** 414 paired reflections let one test whether models know when they were wrong [13, 14] and whether self-reports predict error, using our `outcome_vs_prediction` labels against realized correctness. **(iv) Reasoning and evidence use.** Verbatim reasoning with cited URLs supports factor-attribution and source-use studies via the released lexicon. **(v) Temporal robustness and longitudinal tracking.** Because every target post-dates training, the data probes genuine forecasting rather than recall; the thin search-act-reflect protocol re-runs on any future tournament, enabling live, rolling evaluation of successive model releases against a moving, uncontaminated target.

Reference baselines. To make reuse turnkey, the release ships the numbers a new method must beat: the market’s calibrated probabilities (Table 2) as a forecasting baseline, a flat-stake favourite policy (+\$1,041; Fig. 4) as a betting baseline, and each 2026 agent’s own scored policy as a learned-agent baseline. A new method need only report the same T1–T3 metrics on the same 104 matches, so results are comparable by construction and no data re-collection is required. The paired pre/post structure further supports ablations that a static benchmark cannot: withholding the market citation from an agent’s context, re-scoring under a fixed staking rule, or replaying only the knockout stage each reuse the identical ground truth.

9 Ethics

All betting in this study is virtual; the dataset is a measurement instrument, not gambling advice, and we report profit only to score decision quality. No human subjects or personal data are involved: the records are model outputs and public fixtures, scores, and odds. Odds come from public preview articles with source URLs; we take no book’s proprietary feed. A foreseeable misuse, treating any agent’s edge as an actionable betting strategy, is unsupported by our data, which show no agent beats the market. We disclose that generative AI assisted the parsing and drafting of this work; all findings were verified against the released data by the authors.

10 Limitations

This is a single tournament (104 matches); per-agent differences have wide intervals and we stress patterns over precise rankings. Agents were queried through consumer interfaces with tools enabled, so retrieval and decoding are not fully controlled and providers may update models mid-window, so we hold the user-visible model fixed and record it. Odds are opening lines from mostly one book, not closing consensus. Reflections are the model’s own free text, coded by a transparent but lexical scheme. These are the natural costs of measuring deployed agents on real, unrepeatable events, and the released pipeline makes every choice auditable.

The market baseline is itself a single-book opening line rather than a closing consensus or a betting-exchange midpoint, so it is a conservative, not a maximal, human benchmark; sharper prices would only raise the bar the agents must clear. Our factor coding is lexical and English-only, and although the four-way score cross-check found zero disagreements, the reflection labels are not adjudicated by multiple independent raters. We treat these as documented, auditable simplifications rather than threats to the qualitative claims, each of which rests on gaps far larger than plausible measurement noise.

11 Conclusion

WC2026-Agents turns a future World Cup into a contamination-free benchmark of LLMs as forecasting agents, paired with the betting market as a fifth, economically grounded competitor. Its reference evaluation shows that frontier agents predict alike and cannot beat the market, yet diverge sharply in calibration, betting discipline, and the honesty of their self-reflection, the axes on which they actually differ. We release the data, odds, and evaluation suite to support live, rolling evaluation of agentic forecasting.

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A Reproducibility

One command (`python src/run_all.py`) rebuilds every processed table, analysis CSV, and figure from the raw transcripts and odds; the paper’s numbers are read directly from those outputs. The repository documents the schema, the 13-category reasoning lexicon, per-match ground truth with penalty and advancement fields, and all parse-issue flags. Randomness is not used in evaluation. The four-way score cross-check and the odds/citation validation are part of the build and re-run on every invocation. The full 13×4 factor table, the reliability bins behind Fig. 3, the contrarian breakdown behind Fig. 5, and the match-level upset details are released as CSVs under `data/analysis/`.